

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

COST EFFECTIVE HOUSING TECHNIQUES

(Common to EEE, ME, ECE, CSE, IT, FT, AI&DS, CSE(AI), CSE(DS), CSE(AI&ML), CSE(IoT), CSE(CS), CS&D and AI&ML)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) What is meant by residential density? | 2M |
| (b) What are the various initiatives taken by government to address the urban poor? | 2M |
| (c) How GIS play major role in slum housing projects? | 2M |
| (d) Explain economical wall system in urban housing. | 2M |
| (e) Give a comments on low cost sanitation? | 2M |
| (f) State the applications of Ferro cement? | 2M |
| (g) What are the characteristics of Mud? | 2M |
| (h) Explain any one rural housing programmes. | 2M |
| (i) Specify the earthquake damages to houses. | 2M |
| (j) What are the repairs of earthquake damaged non-engineered buildings? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Explain the status of rural housing finance in India. 10M
- OR**
- 3 (a) Briefly explain the basic concepts and standards for housing programmes. 5M
(b) What is the role of non-government organisations in slum housing? Explain. 5M
- 4 What is prefabrication? Explain few techniques that can be applied in roof construction. 10M
- OR**
- 5 Write a short notes on: 10M
(i) Economic wall system,
(ii) Fly ash gypsum thick for masonry.
- 6 (a) What is Ferro cement? How it could be applied for an alternate building material? 5M
(b) How the industrial wastes are being used for alternate building material? Justify. 5M
- OR**
- 7 Explain in detail about the low-cost sanitation and domestic wall services in low cost infrastructure with their implementation standards as mentioned in IS codes. 10M
- 8 Illustrate the advantages of Mud housing technology with their impacts on plinth level during disaster and suggest for some improvement techniques in using Mud for construction. 10M
- OR**
- 9 (a) Explain fire treatment of thatch roof. 5M
(b) Explain any two soil stabilisation techniques. 5M
- 10 (a) Suggest suitable measures for housing in disaster prone areas. 5M
(b) What are the type of damages and repairs of non-engineered buildings? 5M
- OR**
- 11 Explain the repair and rehabilitation techniques used widely to resist the crack formation in the building during earthquake. 10M

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Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|--|----|
| (a) What is the status of urban housing scenario? | 2M |
| (b) Name any two non-government organizations in slum housing. | 2M |
| (c) What are precast elements in low cost housing? | 2M |
| (d) List the adoption of innovative cost effective techniques in low cost housing. | 2M |
| (e) Differentiate between precast roofing and flooring systems. | 2M |
| (f) List the industrial waste materials. | 2M |
| (g) What are the characteristics of mud? | 2M |
| (h) Justify the fire treatment for thatched roof. | 2M |
| (i) Differentiate between cyclone and flood. | 2M |
| (j) Mention the traditional earthquake prone areas. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- | | | |
|-----------|--|-----|
| 2 | (a) Discuss rural housing status scenario in India. | 5M |
| | (b) Explain various housing finance systems in India. | 5M |
| OR | | |
| 3 | Write a short note on: | 10M |
| | (i) Efficiency of building bye laws, | |
| | (ii) Living Conditions in slums. | |
| 4 | Write a short note on: | 10M |
| | (i) Fly ash, Gypsum brick masonry, | |
| | (ii) Stone block masonry. | |
| OR | | |
| 5 | (a) Explain various precast roofing systems with neat sketches. | 5M |
| | (b) Discuss popular prefabrication systems adopted for mass housing programmes in India. | 5M |
| 6 | (a) Discuss various alternate building materials for low cost housing. | 5M |
| | (b) Discuss low-cost infrastructure services for rural houses. | 5M |
| OR | | |
| 7 | What are future strategies for promotion of alternate construction materials for low-cost housing? | 10M |
| 8 | Discuss how you can provide safe and cost-effective sanitation facilities for Rural housing projects. | 10M |
| OR | | |
| 9 | (a) What are various traditional practices of rural housing? | 5M |
| | (b) Give detailed account on soil stabilization techniques. | 5M |
| 10 | Explain how the floods are mitigated in rural areas. | 10M |
| OR | | |
| 11 | Discuss in detail about the requirements of structural safety of thin pre-cast roofing units against seismic forces. | 10M |
